# An Introduction to Feynman's Technique for Parametric Integration

Applications and Mathematical Justification

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## 1 Introduction

Integration is one of the central operations of calculus. Although elementary techniques such as substitution, integration by parts, and partial fraction decomposition allow us to evaluate many integrals, there are numerous integrals arising in physics and applied mathematics for which these methods are either inefficient or do not lead to an elementary antiderivative.

In some situations, however, an apparently difficult integral becomes considerably easier when we introduce an additional parameter and study how the resulting integral changes with respect to that parameter.

This observation is the central idea behind *parametric integration*, commonly known as *Feynman's technique of differentiating under the integral sign.*

The method is particularly useful when differentiating the integrand with respect to a parameter simplifies its algebraic structure.

To illustrate the general idea, suppose that we wish to evaluate an integral of the form

$$I = \int_a^b f(x)\,dx. \tag{1}$$

Instead of attempting to evaluate (1) directly, we introduce a parameter $\alpha$ and construct a family of integrals

$$F(\alpha) = \int_a^b f(x,\alpha)\,dx, \tag{2}$$

where the original integral is recovered at some particular value $\alpha = \alpha_0$:

$$I = F(\alpha_0).$$

The key observation is that, under suitable mathematical conditions, differentiation with respect to $\alpha$ can be performed inside the integral:

$$\boxed{\frac{d}{d\alpha}\int_a^b f(x,\alpha)\,dx = \int_a^b \frac{\partial f(x,\alpha)}{\partial \alpha}\,dx} \tag{3}$$

The expression on the right-hand side may be substantially easier to evaluate than the original integral.

Once we determine $F'(\alpha)$, we can recover $F(\alpha)$ by integration with respect to $\alpha$. Any resulting constant of integration can then be determined from a known value of $F(\alpha)$ at a convenient parameter, or through a suitable limiting argument.

Finally, evaluating $F$ at $\alpha_0$ recovers the desired integral.

The essential strategy is therefore not to attack a difficult integral directly, but to embed it in a family of integrals whose dependence on a parameter is easier to analyze.

However, an important mathematical question remains. The identity (3) is not universally valid for arbitrary functions. Differentiation and integration cannot always be interchanged.

Why, then, does Feynman's technique work?

We will postpone this question until after developing several applications. We will then investigate the mathematical justification through Leibniz's integral rule and, finally, interpret the technique using the language of linear operators.

# 2 Applications of Feynman's Technique

In this section, we will solve several integral problems using Feynman's technique of differentiating under the integral sign. Our examples will include Gaussian integrals, trigonometric integrals, and logarithmic integrals, some of which are difficult to evaluate using conventional methods.

For the moment, we ask the reader to concentrate on the mechanics of the method: how to introduce an appropriate parameter, how differentiation with respect to that parameter simplifies the integrand, and how the resulting expression leads to the value of the original integral.

We will temporarily assume that differentiation and integration can be interchanged whenever we apply the method. The mathematical conditions under which this interchange is justified will be discussed in a later section.

Our immediate objective is to develop an intuitive and practical understanding of the technique through explicit calculations.

## 2.1 Gaussian Integrals: Odd Powers

Let us begin by considering the family of Gaussian integrals

$$I_n = \int_0^\infty x^n e^{-x^2}\, dx, \qquad n = 0, 1, 2, \ldots \tag{4}$$

These integrals arise frequently in probability theory, statistics, and mathematical physics.

We will treat odd and even values of $n$ separately, since the corresponding calculations require slightly different approaches.

First, suppose that $n$ is odd. We may write

$$n = 2m + 1, \qquad m = 0, 1, 2, \ldots$$

Our objective is therefore to evaluate

$$I_{2m+1} = \int_0^\infty x^{2m+1} e^{-x^2}\, dx. \tag{5}$$

**Step 1: A preliminary substitution**

Before introducing a parameter, we make the substitution

$$u = x^2, \qquad du = 2x\, dx.$$

Since

$$x^{2m+1}\, dx = (x^2)^m x\, dx = \frac{1}{2} u^m\, du,$$

equation (5) becomes

$$I_{2m+1} = \frac{1}{2} \int_0^\infty u^m e^{-u}\, du. \tag{6}$$

Although the substitution has simplified the exponential, we still have the factor $u^m$ inside the integral.

This is where Feynman's technique becomes useful.

**Step 2: Introducing a parameter**

Consider the parameter-dependent integral

$$F(\alpha) = \int_0^\infty e^{-\alpha u}\, du, \qquad \alpha > 0. \tag{7}$$

The reason for introducing the parameter in the exponent is that

$$\frac{\partial}{\partial \alpha} e^{-\alpha u} = -u e^{-\alpha u}.$$

Differentiating a second time gives

$$\frac{\partial^2}{\partial \alpha^2} e^{-\alpha u} = u^2 e^{-\alpha u}.$$

More generally,

$$\frac{\partial^m}{\partial \alpha^m} e^{-\alpha u} = (-1)^m u^m e^{-\alpha u}. \tag{8}$$

Thus, repeated differentiation with respect to the parameter generates precisely the powers of $u$ that appear in our original integral.

**Step 3: Applying Feynman's technique**

Differentiating (7) $m$ times with respect to $\alpha$, we obtain

$$\begin{aligned} F^{(m)}(\alpha) &= \frac{d^m}{d\alpha^m} \int_0^\infty e^{-\alpha u}\, du \\ &= \int_0^\infty \frac{\partial^m}{\partial \alpha^m} e^{-\alpha u}\, du \\ &= (-1)^m \int_0^\infty u^m e^{-\alpha u}\, du. \end{aligned} \tag{9}$$

Consequently,

$$\int_0^\infty u^m e^{-\alpha u}\, du = (-1)^m F^{(m)}(\alpha). \tag{10}$$

Comparing this expression with (6), we see that the desired integral can be recovered by setting $\alpha = 1$.

Therefore,

$$I_{2m+1} = \frac{(-1)^m}{2}\, F^{(m)}(\alpha)\Big|_{\alpha=1}. \tag{11}$$

**Step 4: Evaluating the parametric integral**

The integral defining $F(\alpha)$ is elementary:

$$\begin{aligned} F(\alpha) &= \int_0^\infty e^{-\alpha u}\, du \\ &= \left[-\frac{1}{\alpha} e^{-\alpha u}\right]_0^\infty \\ &= \frac{1}{\alpha}. \end{aligned} \tag{12}$$

Hence,

$$F(\alpha) = \alpha^{-1}.$$

By repeated differentiation,

$$F^{(m)}(\alpha) = (-1)^m m!\, \alpha^{-(m+1)}. \tag{13}$$

Substituting into (11), we obtain

$$\begin{aligned} I_{2m+1} &= \frac{(-1)^m}{2} \left[(-1)^m m!\, \alpha^{-(m+1)}\right]_{\alpha=1} \\ &= \frac{m!}{2}. \end{aligned} \tag{14}$$

We have therefore established

$$\boxed{\int_0^\infty x^{2m+1} e^{-x^2}\, dx = \frac{m!}{2}, \qquad m = 0, 1, 2, \ldots} \tag{15}$$

Notice the essential mechanism behind the calculation. Rather than attempting to integrate $u^m e^{-u}$ directly, we introduced a parameter into the exponential function. Differentiation with respect to that parameter generated the required power of $u$, while the parametric integral itself remained elementary.

This is the principal idea underlying Feynman's technique.

## 2.2 Gaussian Integrals: Even Powers

We now turn our attention to Gaussian integrals involving even powers of $x$. Our objective is to evaluate

$$I_{2m} = \int_0^\infty x^{2m} e^{-x^2}\, dx, \qquad m = 0, 1, 2, \ldots \tag{16}$$

Unlike the odd-power case, we will not make an initial substitution. Instead, we introduce a parameter directly into the exponent.

### Step 1: Introducing a parameter

Consider the parametric integral

$$G(\alpha) = \int_0^\infty e^{-\alpha x^2}\, dx, \qquad \alpha > 0. \tag{17}$$

The reason for introducing $\alpha$ into the exponent is that

$$\frac{\partial}{\partial \alpha} e^{-\alpha x^2} = -x^2 e^{-\alpha x^2}.$$

Differentiating again gives

$$\frac{\partial^2}{\partial \alpha^2} e^{-\alpha x^2} = x^4 e^{-\alpha x^2}.$$

Continuing this process, we obtain

$$\frac{\partial^m}{\partial \alpha^m} e^{-\alpha x^2} = (-1)^m x^{2m} e^{-\alpha x^2}. \tag{18}$$

Thus, each differentiation with respect to $\alpha$ produces an additional factor of $-x^2$.

Applying Feynman's technique,

$$
\begin{aligned}
G^{(m)}(\alpha) &= \frac{d^m}{d\alpha^m} \int_0^\infty e^{-\alpha x^2}\, dx \\
&= \int_0^\infty \frac{\partial^m}{\partial \alpha^m} e^{-\alpha x^2}\, dx \\
&= (-1)^m \int_0^\infty x^{2m} e^{-\alpha x^2}\, dx. \qquad (19)
\end{aligned}
$$

Therefore,

$$
I_{2m} = (-1)^m \, G^{(m)}(\alpha)\Big|_{\alpha=1}. \qquad (20)
$$

We have now reduced the problem to finding the derivatives of $G(\alpha)$.

**Step 2: Evaluating the parametric integral**

Recall the classical Gaussian integral

$$
\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}. \qquad (21)
$$

Since $e^{-x^2}$ is an even function, it follows that

$$
\int_0^\infty e^{-x^2}\, dx = \frac{\sqrt{\pi}}{2}. \qquad (22)
$$

We can now evaluate $G(\alpha)$ by making the substitution

$$
u = \sqrt{\alpha}\, x, \qquad dx = \frac{du}{\sqrt{\alpha}}.
$$

Consequently,

$$
\begin{aligned}
G(\alpha) &= \int_0^\infty e^{-\alpha x^2}\, dx \\
&= \frac{1}{\sqrt{\alpha}} \int_0^\infty e^{-u^2}\, du \\
&= \frac{\sqrt{\pi}}{2\sqrt{\alpha}}. \qquad (23)
\end{aligned}
$$

Hence,

$$
\boxed{G(\alpha) = \frac{\sqrt{\pi}}{2} \alpha^{-1/2}} \qquad (24)
$$

**Step 3: Repeated differentiation**

We now need to compute the $m$-th derivative of $\alpha^{-1/2}$.

The first derivative is

$$\frac{d}{d\alpha}\alpha^{-1/2} = -\frac{1}{2}\alpha^{-3/2}.$$

Differentiating a second time,

$$\frac{d^2}{d\alpha^2}\alpha^{-1/2} = \frac{1\cdot 3}{2^2}\alpha^{-5/2}.$$

The third derivative gives

$$\frac{d^3}{d\alpha^3}\alpha^{-1/2} = -\frac{1\cdot 3\cdot 5}{2^3}\alpha^{-7/2}.$$

Notice the pattern. Every differentiation introduces another negative sign and another odd integer in the numerator.

After $m$ differentiations,

$$\frac{d^m}{d\alpha^m}\alpha^{-1/2} = (-1)^m \frac{1\cdot 3\cdot 5\cdots(2m-1)}{2^m}\alpha^{-(m+\frac{1}{2})}. \tag{25}$$

The product

$$1\cdot 3\cdot 5\cdots(2m-1)$$

is called the *double factorial* of $2m-1$, denoted by

$$(2m-1)!!.$$

It is important to distinguish the double factorial from the ordinary factorial.

For example,

$$5! = 5\cdot 4\cdot 3\cdot 2\cdot 1 = 120,$$

whereas

$$5!! = 5\cdot 3\cdot 1 = 15.$$

Thus, equation (25) can be written as

$$\frac{d^m}{d\alpha^m}\alpha^{-1/2} = (-1)^m \frac{(2m-1)!!}{2^m}\alpha^{-(m+1/2)}. \tag{26}$$

Substituting into (24),

$$G^{(m)}(\alpha) = \frac{\sqrt{\pi}}{2}(-1)^m \frac{(2m-1)!!}{2^m}\alpha^{-(m+1/2)}. \tag{27}$$

Finally, using (20) and setting $\alpha = 1$,

$$\begin{aligned} I_{2m} &= (-1)^m G^{(m)}(1) \\ &= (-1)^m \frac{\sqrt{\pi}}{2} (-1)^m \frac{(2m-1)!!}{2^m} \\ &= \frac{\sqrt{\pi}(2m-1)!!}{2^{m+1}}. \end{aligned} \tag{28}$$

We have therefore obtained

$$\boxed{\int_0^\infty x^{2m} e^{-x^2}\, dx = \frac{\sqrt{\pi}(2m-1)!!}{2^{m+1}}} \tag{29}$$

Here we use the convention $(-1)!! = 1$ when $m = 0$, so that the formula includes the standard Gaussian integral.

**Step 4: Expressing the double factorial in terms of ordinary factorials**

Although the double factorial arises naturally through repeated differentiation, it can also be expressed using ordinary factorials.

To see this, consider

$$(2m)! = 1 \cdot 2 \cdot 3 \cdots (2m).$$

We can separate this product into its odd and even factors:

$$\begin{aligned} (2m)! = &\underbrace{(1 \cdot 3 \cdot 5 \cdots (2m-1))}_{\text{Odd factors}} \\ &\times \underbrace{(2 \cdot 4 \cdot 6 \cdots 2m)}_{\text{Even factors}}. \end{aligned} \tag{30}$$

The first product is $(2m-1)!!$.

For the second product, we can factor out 2 from each of its $m$ terms:

$$\begin{aligned} 2 \cdot 4 \cdot 6 \cdots (2m) &= (2 \cdot 1)(2 \cdot 2) \cdots (2 \cdot m) \\ &= 2^m (1 \cdot 2 \cdots m) \\ &= 2^m m!. \end{aligned} \tag{31}$$

Therefore,

$$(2m)! = (2m-1)!!\, 2^m m!.$$

Rearranging,

$$\boxed{(2m-1)!! = \frac{(2m)!}{2^m m!}} \tag{32}$$

Substituting this identity into (29), we obtain

$$\begin{aligned} I_{2m} &= \frac{\sqrt{\pi}}{2^{m+1}} \frac{(2m)!}{2^m m!} \\ &= \frac{\sqrt{\pi}(2m)!}{2^{2m+1} m!}. \end{aligned} \tag{33}$$

Thus, an equivalent expression for our Gaussian integral is

$$\boxed{\int_0^\infty x^{2m} e^{-x^2}\, dx = \frac{\sqrt{\pi}(2m)!}{2^{2m+1} m!}, \qquad m = 0, 1, 2, \ldots} \tag{34}$$

We have now evaluated the Gaussian integrals for both odd and even powers. For convenience, we summarize the results:

$$\boxed{\begin{aligned} &\int_0^\infty x^{2m+1} e^{-x^2}\, dx = \frac{m!}{2}, \\ &\int_0^\infty x^{2m} e^{-x^2}\, dx = \frac{\sqrt{\pi}(2m)!}{2^{2m+1} m!}. \end{aligned}} \tag{35}$$

The two calculations illustrate the same underlying principle: by introducing a parameter into an exponential function, powers of the integration variable can be generated through repeated differentiation.

## 2.3 The Dirichlet Integral

We now consider a famous improper integral, commonly known as the *Dirichlet integral*:

$$I = \int_0^\infty \frac{\sin x}{x}\, dx. \tag{36}$$

Unlike the Gaussian integrals considered previously, this integral involves an oscillatory function divided by $x$. Moreover, the function $\sin x/x$ does not have an elementary antiderivative.

Nevertheless, Feynman's technique allows us to evaluate the definite integral through a relatively straightforward calculation.

The central idea is to introduce a parameter in such a way that differentiation removes the factor $1/x$ from the integrand.

### Step 1: Introducing a parameter

Consider the parametric integral

$$F(\alpha) = \int_0^\infty \frac{\sin x}{x} e^{-\alpha x}\, dx, \qquad \alpha > 0. \tag{37}$$

Notice that we have multiplied the original integrand by an exponential factor $e^{-\alpha x}$.

This factor serves two purposes.

First, for $\alpha > 0$, the exponential decays rapidly as $x \to \infty$, making the integral absolutely convergent.

Second, differentiating $e^{-\alpha x}$ with respect to $\alpha$ produces a factor of $-x$, which will cancel the denominator.

The original integral will eventually be recovered by allowing $\alpha \to 0^+$.

**Step 2: Differentiating under the integral sign**

Differentiating $F(\alpha)$, we obtain

$$\begin{aligned} F'(\alpha) &= \frac{d}{d\alpha} \int_0^\infty \frac{\sin x}{x} e^{-\alpha x}\, dx \\ &= \int_0^\infty \frac{\partial}{\partial \alpha} \left( \frac{\sin x}{x} e^{-\alpha x} \right) dx \\ &= \int_0^\infty \frac{\sin x}{x} (-x) e^{-\alpha x}\, dx. \end{aligned} \tag{38}$$

The factor $x$ in the numerator cancels with the denominator, leaving

$$\boxed{F'(\alpha) = -\int_0^\infty e^{-\alpha x} \sin x\, dx.} \tag{39}$$

This is the crucial simplification. The original integral contained the troublesome factor $1/x$, but differentiation with respect to the parameter has eliminated it.

We are now left with an elementary exponential-trigonometric integral.

**Step 3: Evaluating the resulting integral**

To evaluate the integral on the right-hand side of (39), we use Euler's identity

$$e^{ix} = \cos x + i \sin x,$$

where $i^2 = -1$.

In particular,

$$\sin x = \mathrm{Im}(e^{ix}),$$

where Im denotes the imaginary part of a complex number.

Therefore,

$$\begin{aligned} \int_0^\infty e^{-\alpha x} \sin x\, dx &= \mathrm{Im} \left( \int_0^\infty e^{-\alpha x} e^{ix}\, dx \right) \\ &= \mathrm{Im} \left( \int_0^\infty e^{(i-\alpha)x}\, dx \right). \end{aligned} \tag{40}$$

The integral inside the imaginary-part operator is elementary:

$$\begin{aligned}\int_0^\infty e^{(i-\alpha)x}\,dx &= \left[\frac{e^{(i-\alpha)x}}{i-\alpha}\right]_0^\infty \\ &= -\frac{1}{i-\alpha} \\ &= \frac{1}{\alpha-i}.\end{aligned} \tag{41}$$

At this point, it is worth explaining why the exponential term vanishes at infinity.

Observe that

$$e^{(i-\alpha)x} = e^{ix}e^{-\alpha x}.$$

Since

$$|e^{ix}| = 1,$$

we have

$$|e^{(i-\alpha)x}| = e^{-\alpha x}. \tag{42}$$

For $\alpha > 0$,

$$\lim_{x\to\infty} e^{-\alpha x} = 0.$$

Consequently,

$$\lim_{x\to\infty} e^{(i-\alpha)x} = 0. \tag{43}$$

Thus, although $e^{ix}$ oscillates, the exponential damping factor forces the entire expression to approach zero.

Returning to our calculation, we obtain

$$\begin{aligned}\frac{1}{\alpha-i} &= \frac{\alpha+i}{(\alpha-i)(\alpha+i)} \\ &= \frac{\alpha+i}{\alpha^2+1}.\end{aligned} \tag{44}$$

Taking the imaginary part gives

$$\int_0^\infty e^{-\alpha x}\sin x\,dx = \frac{1}{1+\alpha^2}. \tag{45}$$

Therefore, equation (39) becomes

$$\boxed{F'(\alpha) = -\frac{1}{1+\alpha^2}.} \tag{46}$$

**Step 4: Recovering the parametric integral**

We now have a simple differential equation for $F(\alpha)$:

$$\frac{dF}{d\alpha} = -\frac{1}{1+\alpha^2}.$$

Integrating with respect to $\alpha$, we obtain

$$F(\alpha) = -\arctan(\alpha) + C, \tag{47}$$

where $C$ is a constant of integration.

To determine $C$, we examine the behavior of $F(\alpha)$ as $\alpha \to \infty$.

Recall that

$$F(\alpha) = \int_0^\infty \frac{\sin x}{x} e^{-\alpha x}\, dx.$$

Since

$$\left|\frac{\sin x}{x}\right| \leq 1, \qquad x > 0,$$

we obtain

$$\begin{aligned} |F(\alpha)| &\leq \int_0^\infty \left|\frac{\sin x}{x}\right| e^{-\alpha x}\, dx \\ &\leq \int_0^\infty e^{-\alpha x}\, dx \\ &= \frac{1}{\alpha}. \end{aligned} \tag{48}$$

Therefore,

$$\lim_{\alpha\to\infty} F(\alpha) = 0.$$

Applying this limit to (47),

$$0 = -\frac{\pi}{2} + C.$$

Hence,

$$C = \frac{\pi}{2}.$$

We have consequently obtained

$$\boxed{F(\alpha) = \frac{\pi}{2} - \arctan(\alpha), \qquad \alpha > 0.} \tag{49}$$

**Step 5: Recovering the original integral**

We now return to our original objective:

$$I = \int_0^\infty \frac{\sin x}{x}\, dx.$$

Recall that the parameter $\alpha$ was introduced only as an auxiliary variable.

The original integrand is recovered when the damping factor $e^{-\alpha x}$ approaches 1.

Thus, we consider

$$\lim_{\alpha \to 0^+} F(\alpha).$$

Using (49),

$$\begin{aligned} \lim_{\alpha \to 0^+} F(\alpha) &= \lim_{\alpha \to 0^+} \left( \frac{\pi}{2} - \arctan \alpha \right) \\ &= \frac{\pi}{2}. \end{aligned} \tag{50}$$

The original improper integral converges by Dirichlet's test. Moreover, the exponential damping preserves its value in the limit $\alpha \to 0^+$. This limiting argument will be discussed when we examine the mathematical justification of the method.

We therefore conclude that

$$\boxed{\int_0^\infty \frac{\sin x}{x}\, dx = \frac{\pi}{2}.} \tag{51}$$

Notice how the choice of parameter determined the success of the method. By introducing an exponential damping factor, we transformed an integral involving $\sin x/x$ into an elementary integral involving $e^{-\alpha x} \sin x$.

In contrast with the Gaussian examples, where repeated differentiation generated powers of the integration variable, here a single differentiation eliminated an unwanted factor from the denominator.

This is another important application of Feynman's technique.

## 2.4 Gaussian–Cosine Integral: A Cauchy Problem

In our previous examples, differentiating under the integral sign either simplified the integrand directly or produced an elementary expression for the derivative of the parametric integral.

There are, however, situations in which differentiation with respect to a parameter produces another integral that is not immediately recognizable.

In such cases, a familiar technique such as integration by parts may allow us to express the new integral in terms of the original parametric integral.

This leads to an interesting possibility: instead of evaluating the integral directly, we may derive an ordinary differential equation satisfied by it.

If the value of the parametric integral is known at a convenient parameter, we obtain an initial value problem, also called a *Cauchy problem*.

We can then determine the original integral by solving the resulting differential equation.

To illustrate this connection between Feynman's technique and ordinary differential equations, consider

$$I(\alpha) = \int_0^\infty e^{-x^2} \cos(\alpha x)\, dx, \qquad \alpha \in \mathbb{R}. \tag{52}$$

This integral is closely related to the Fourier transform of a Gaussian function, an important object in probability theory, signal processing, and mathematical physics.

Our objective is to obtain an explicit expression for $I(\alpha)$.

**Step 1: Differentiating under the integral sign**

We begin by applying Feynman's technique.

Differentiating with respect to $\alpha$, we obtain

$$\begin{aligned} I'(\alpha) &= \frac{d}{d\alpha} \int_0^\infty e^{-x^2} \cos(\alpha x)\, dx \\ &= \int_0^\infty \frac{\partial}{\partial \alpha} \left( e^{-x^2} \cos(\alpha x) \right) dx \\ &= -\int_0^\infty x e^{-x^2} \sin(\alpha x)\, dx. \end{aligned} \tag{53}$$

Notice that differentiation has produced a factor of $x$, together with a sine function.

Unlike our earlier Gaussian examples, the resulting integral is not immediately elementary.

However, we should recognize an important relationship:

$$\frac{d}{dx} e^{-x^2} = -2x e^{-x^2}. \tag{54}$$

Therefore,

$$x e^{-x^2} = -\frac{1}{2} \frac{d}{dx} e^{-x^2}.$$

This suggests applying integration by parts to the right-hand side of (53).

As we will see, this will allow us to recover the original integral $I(\alpha)$.

**Step 2: Applying integration by parts**

For convenience, let

$$K(\alpha) = \int_0^\infty x e^{-x^2} \sin(\alpha x)\, dx. \tag{55}$$

Then equation (53) becomes

$$I'(\alpha) = -K(\alpha).$$

To evaluate $K(\alpha)$, we use integration by parts.

Choose

$$u = \sin(\alpha x), \qquad dv = xe^{-x^2}\,dx.$$

Then

$$du = \alpha\cos(\alpha x)\,dx$$

and

$$v = -\frac{1}{2}e^{-x^2}.$$

Recall the integration by parts formula

$$\int u\,dv = uv - \int v\,du.$$

Applying this formula, we obtain

$$\begin{aligned} K(\alpha) = &\left[-\frac{1}{2}e^{-x^2}\sin(\alpha x)\right]_0^\infty \\ &+ \frac{\alpha}{2}\int_0^\infty e^{-x^2}\cos(\alpha x)\,dx. \end{aligned} \tag{56}$$

Let us examine the boundary term.

At $x = 0$, we have

$$e^{-0^2}\sin(\alpha\cdot 0) = 0.$$

At infinity, observe that

$$\left|e^{-x^2}\sin(\alpha x)\right| \le e^{-x^2}.$$

Since

$$\lim_{x\to\infty} e^{-x^2} = 0,$$

the boundary term vanishes at infinity as well.

Consequently,

$$K(\alpha) = \frac{\alpha}{2}\int_0^\infty e^{-x^2}\cos(\alpha x)\,dx. \tag{57}$$

But the integral on the right-hand side is precisely our original parametric integral $I(\alpha)$.

Thus,

$$K(\alpha) = \frac{\alpha}{2} I(\alpha). \tag{58}$$

Recalling that

$$I'(\alpha) = -K(\alpha),$$

we finally obtain

$$\boxed{I'(\alpha) = -\frac{\alpha}{2} I(\alpha).} \tag{59}$$

This is the crucial observation.

By combining differentiation under the integral sign with integration by parts, we have transformed the original integral into a first-order linear homogeneous differential equation.

Unlike the previous examples, the derivative $I'(\alpha)$ is now expressed in terms of $I(\alpha)$ itself.

**Step 3: Formulating the Cauchy problem**

The differential equation (59) can be written as

$$I'(\alpha) + \frac{\alpha}{2} I(\alpha) = 0. \tag{60}$$

To determine a unique solution, we need an initial condition.

A natural choice is $\alpha = 0$, because the cosine term becomes 1.

Indeed, returning to (52),

$$\begin{aligned} I(0) &= \int_0^\infty e^{-x^2} \cos(0)\, dx \\ &= \int_0^\infty e^{-x^2}\, dx. \end{aligned} \tag{61}$$

But this is the standard Gaussian integral.

Recall from (22) that

$$\int_0^\infty e^{-x^2}\, dx = \frac{\sqrt{\pi}}{2}.$$

Therefore,

$$\boxed{I(0) = \frac{\sqrt{\pi}}{2}.} \tag{62}$$

Combining the differential equation with this initial condition, we obtain

$$\boxed{\begin{cases} I'(\alpha) + \dfrac{\alpha}{2} I(\alpha) = 0, \\ I(0) = \dfrac{\sqrt{\pi}}{2}. \end{cases}} \tag{63}$$

This is a Cauchy problem for a first-order linear ordinary differential equation.

Notice that we have not evaluated the original integral directly.

Instead, Feynman's technique has allowed us to formulate a differential equation whose solution will give the value of the integral for every real $\alpha$.

**Step 4: Solving the Cauchy problem**

We now solve

$$I'(\alpha) + \frac{\alpha}{2} I(\alpha) = 0. \tag{64}$$

This is a first-order linear homogeneous differential equation.

We will use the method of integrating factors.

Recall that an equation of the form

$$y'(t) + p(t)y(t) = 0$$

has integrating factor

$$\mu(t) = e^{\int p(t)\, dt}.$$

In our problem,

$$p(\alpha) = \frac{\alpha}{2}.$$

Therefore, we choose

$$\mu(\alpha) = e^{\alpha^2/4}. \tag{65}$$

Multiplying the differential equation by $\mu(\alpha)$ gives

$$e^{\alpha^2/4} I'(\alpha) + \frac{\alpha}{2} e^{\alpha^2/4} I(\alpha) = 0. \tag{66}$$

Now observe that

$$\frac{d}{d\alpha} e^{\alpha^2/4} = \frac{\alpha}{2} e^{\alpha^2/4}.$$

Thus, by the product rule,

$$\frac{d}{d\alpha}\left(e^{\alpha^2/4}I(\alpha)\right)$$
$$= e^{\alpha^2/4}I'(\alpha) + \frac{\alpha}{2}e^{\alpha^2/4}I(\alpha). \tag{67}$$

Consequently, our differential equation becomes

$$\frac{d}{d\alpha}\left(e^{\alpha^2/4}I(\alpha)\right) = 0. \tag{68}$$

A function whose derivative vanishes throughout an interval must be constant on that interval. Therefore,

$$e^{\alpha^2/4}I(\alpha) = C,$$

where $C$ is a constant.

Solving for $I(\alpha)$,

$$I(\alpha) = Ce^{-\alpha^2/4}. \tag{69}$$

We now apply the initial condition obtained in Step 3:

$$I(0) = \frac{\sqrt{\pi}}{2}.$$

Substituting $\alpha = 0$ into (69),

$$\frac{\sqrt{\pi}}{2} = Ce^0 = C.$$

Hence,

$$C = \frac{\sqrt{\pi}}{2}.$$

We have therefore obtained

$$I(\alpha) = \frac{\sqrt{\pi}}{2}e^{-\alpha^2/4}. \tag{70}$$

Since the coefficient $\alpha/2$ is continuous on $\mathbb{R}$, the standard existence and uniqueness theorem for first-order linear ODEs guarantees that this is the unique solution of the Cauchy problem.

Returning to the definition of $I(\alpha)$, we conclude that

$$\boxed{\int_0^\infty e^{-x^2}\cos(\alpha x)\,dx = \frac{\sqrt{\pi}}{2}e^{-\alpha^2/4}, \qquad \alpha \in \mathbb{R}.} \tag{71}$$

The result is remarkable. An improper integral involving an oscillatory trigonometric function has been evaluated by solving a simple initial value problem.

The essential steps were to differentiate with respect to the parameter, apply integration by parts, recognize the original integral, and use a convenient parameter value to obtain an initial condition.

## 2.5 A Logarithmic Integral

Our final example concerns a family of logarithmic integrals.

Consider

$$I(\alpha) = \int_0^1 \frac{x^\alpha - 1}{\ln x}\, dx, \qquad \alpha > -1. \tag{72}$$

The presence of $\ln x$ in the denominator makes direct integration difficult.

Furthermore, the integrand appears to be undefined at $x = 1$, since both the numerator and denominator vanish there.

However, this is only a removable singularity. Indeed,

$$\lim_{x\to 1} \frac{x^\alpha - 1}{\ln x} = \alpha.$$

Thus, the behavior near $x = 1$ causes no difficulty.

Unlike our previous examples, the parameter is already present in the integral. We therefore do not need to introduce an additional one.

### Step 1: Differentiating with respect to the parameter

We begin by differentiating $I(\alpha)$:

$$\begin{aligned} I'(\alpha) &= \frac{d}{d\alpha} \int_0^1 \frac{x^\alpha - 1}{\ln x}\, dx \\ &= \int_0^1 \frac{\partial}{\partial \alpha} \left( \frac{x^\alpha - 1}{\ln x} \right) dx. \end{aligned} \tag{73}$$

To compute the derivative inside the integral, recall that

$$x^\alpha = e^{\alpha \ln x}.$$

Consequently,

$$\frac{\partial}{\partial \alpha} x^\alpha = x^\alpha \ln x. \tag{74}$$

Therefore,

$$\begin{aligned} I'(\alpha) &= \int_0^1 \frac{x^\alpha \ln x}{\ln x}\, dx \\ &= \int_0^1 x^\alpha\, dx. \end{aligned} \tag{75}$$

We have obtained

$$\boxed{I'(\alpha) = \int_0^1 x^\alpha \, dx.} \tag{76}$$

Notice what happened: differentiating $x^\alpha$ produced precisely the factor $\ln x$ required to cancel the denominator.

This is the essential idea behind our choice of differentiation.

**Step 2: Evaluating the resulting integral**

The integral on the right-hand side of (76) is now elementary.

For $\alpha > -1$,

$$\begin{aligned} I'(\alpha) &= \int_0^1 x^\alpha \, dx \\ &= \left[ \frac{x^{\alpha+1}}{\alpha+1} \right]_0^1 \\ &= \frac{1}{\alpha+1}. \end{aligned} \tag{77}$$

Thus,

$$\boxed{I'(\alpha) = \frac{1}{\alpha+1}.} \tag{78}$$

**Step 3: Recovering the original integral**

We now integrate with respect to $\alpha$:

$$\frac{dI}{d\alpha} = \frac{1}{\alpha+1}.$$

Therefore,

$$I(\alpha) = \ln(\alpha+1) + C, \tag{79}$$

where $C$ is a constant of integration.

To determine this constant, we use a convenient value of the parameter.

Setting $\alpha = 0$ in the original integral gives

$$\begin{aligned} I(0) &= \int_0^1 \frac{x^0 - 1}{\ln x} \, dx \\ &= \int_0^1 0 \, dx \\ &= 0. \end{aligned} \tag{80}$$

On the other hand, equation (79) gives

$$I(0) = \ln 1 + C = C.$$

Consequently,

$$C = 0.$$

We have therefore established

$$\boxed{\int_0^1 \frac{x^\alpha - 1}{\ln x}\, dx = \ln(\alpha + 1), \qquad \alpha > -1.} \tag{81}$$

As a particular example, setting $\alpha = 1$ gives

$$\boxed{\int_0^1 \frac{x - 1}{\ln x}\, dx = \ln 2.} \tag{82}$$

This example is particularly instructive because the parameter was already present in the original integral.

Differentiation generated a logarithmic factor that canceled the denominator, reducing the problem to an elementary power integral.

The constant of integration was then determined by evaluating the original parametric integral at a convenient parameter value.

### 2.6 Observations on the Method

The preceding examples illustrate several ways in which Feynman's technique can simplify difficult integrals.

In the Gaussian integrals, differentiation with respect to a parameter generated powers of the integration variable.

In the Dirichlet integral, differentiation removed an unwanted factor of $1/x$.

In the Gaussian–cosine integral, differentiation followed by integration by parts produced an ordinary differential equation. A known initial condition then allowed us to formulate and solve a Cauchy problem.

In the logarithmic integral, differentiation generated a factor of $\ln x$ that canceled the denominator.

Although the calculations look different, the common principle remains the same: we replace a difficult integration problem with the study of a suitable parameter-dependent integral.

Once the derivative with respect to the parameter has been evaluated, the original integral can often be recovered through elementary integration and an appropriate boundary or limiting condition.

At this point, however, an important mathematical question remains unanswered.

Throughout these examples, we have interchanged differentiation and integration without establishing the conditions under which this operation is legitimate.

We now turn to the mathematical foundation of Feynman's technique, beginning with Leibniz's integral rule and eventually examining the problem from the perspective of linear operators.

# 3 Why Does Feynman's Technique Work?

In the preceding section, we evaluated several nontrivial integrals using Feynman's technique of differentiating under the integral sign.

Although the calculations were relatively straightforward, we repeatedly used an important identity without establishing its mathematical validity:

$$\frac{d}{d\alpha}\int_a^b f(x,\alpha)\,dx = \int_a^b \frac{\partial f(x,\alpha)}{\partial \alpha}\,dx. \tag{83}$$

The left-hand side tells us to integrate first and then differentiate, while the right-hand side reverses the order of these operations.

Why should these two procedures produce the same result?

To understand the underlying idea, we begin with the definition of integration through Riemann sums.

## 3.1 An Intuitive Explanation Using Riemann Sums

Recall from elementary calculus that a definite integral can be defined as the limit of a sequence of Riemann sums.

Suppose that $f(x,\alpha)$ is a sufficiently regular function on the interval $[a,b]$.

We divide $[a,b]$ into $n$ subintervals,

$$a = x_0 < x_1 < \cdots < x_n = b,$$

and let

$$\Delta x_i = x_i - x_{i-1}.$$

Choosing a point $\xi_i$ from each subinterval, we construct the Riemann sum

$$S_n(\alpha) = \sum_{i=1}^{n} f(\xi_i,\alpha)\Delta x_i. \tag{84}$$

As the maximum width of the subintervals approaches zero, the Riemann sums converge to the definite integral:

$$F(\alpha) = \int_a^b f(x,\alpha)\,dx = \lim_{n\to\infty} S_n(\alpha). \tag{85}$$

Now observe that each $S_n(\alpha)$ is simply a finite sum of functions of the parameter $\alpha$.

Since differentiation is a linear operation, we can differentiate this finite sum term by term:

$$S_n'(\alpha) = \frac{d}{d\alpha} \sum_{i=1}^{n} f(\xi_i, \alpha) \Delta x_i$$
$$= \sum_{i=1}^{n} \frac{\partial f(\xi_i, \alpha)}{\partial \alpha} \Delta x_i. \tag{86}$$

Notice that $\Delta x_i$ does not depend on $\alpha$, since the interval of integration is fixed.

The expression on the right-hand side of (86) is itself a Riemann sum, this time for the function

$$\frac{\partial f(x, \alpha)}{\partial \alpha}.$$

Consequently, if we take the limit as the partition becomes finer, we expect

$$\lim_{n \to \infty} S_n'(\alpha) = \int_a^b \frac{\partial f(x, \alpha)}{\partial \alpha} \, dx. \tag{87}$$

On the other hand, since

$$F(\alpha) = \lim_{n \to \infty} S_n(\alpha),$$

we would like to write

$$F'(\alpha) = \lim_{n \to \infty} S_n'(\alpha).$$

Combining these observations would give

$$\boxed{\frac{d}{d\alpha} \int_a^b f(x, \alpha) \, dx = \int_a^b \frac{\partial f(x, \alpha)}{\partial \alpha} \, dx.}$$

This is precisely the identity used in Feynman's technique.

However, there is a subtle mathematical issue hidden in this argument.

Although differentiation can always be distributed over a finite sum, it cannot necessarily be interchanged with an infinite limit.

In particular, the identity

$$\frac{d}{d\alpha} \left( \lim_{n \to \infty} S_n(\alpha) \right) = \lim_{n \to \infty} S_n'(\alpha)$$

requires additional assumptions.

Thus, the Riemann-sum argument provides a useful intuition, but it does not, by itself, constitute a complete proof.

To establish the result rigorously, we turn to Leibniz's integral rule.

## 3.2 Leibniz's Integral Rule

We now establish the mathematical justification for differentiation under the integral sign.

We first consider the case where the limits of integration are fixed, that is, they do not depend on the parameter $\alpha$.

**Theorem 3.1** (Differentiation Under the Integral Sign)**.** *Let $J \subseteq \mathbb{R}$ be an open interval, and let $a < b$. Suppose that $f(x, \alpha)$ and its partial derivative*

$$\frac{\partial f(x, \alpha)}{\partial \alpha}$$

*are continuous on $[a, b] \times J$.*

*Define*

$$F(\alpha) = \int_a^b f(x, \alpha)\, dx.$$

*Then $F$ is differentiable on $J$, and*

$$\boxed{F'(\alpha) = \int_a^b \frac{\partial f(x, \alpha)}{\partial \alpha}\, dx.} \tag{88}$$

*Proof.* We will establish the theorem directly from the definition of the derivative.

**Step 1: Applying the definition of the derivative.**

Fix an arbitrary $\alpha \in J$. By the definition of the derivative,

$$F'(\alpha) = \lim_{h \to 0} \frac{F(\alpha + h) - F(\alpha)}{h}. \tag{89}$$

Recall that

$$F(\alpha) = \int_a^b f(x, \alpha)\, dx.$$

Therefore, for $h \neq 0$,

$$\begin{aligned} &\frac{F(\alpha + h) - F(\alpha)}{h} \\ &= \frac{1}{h}\left[\int_a^b f(x, \alpha + h)\, dx - \int_a^b f(x, \alpha)\, dx\right] \\ &= \int_a^b \frac{f(x, \alpha + h) - f(x, \alpha)}{h}\, dx. \end{aligned} \tag{90}$$

The last equality follows from the linearity of definite integration.

Notice that, at this stage, we have not interchanged differentiation and integration. We have only used the linearity of the integral.

Our objective is to prove that

$$\lim_{h\to 0}\int_a^b \frac{f(x,\alpha+h)-f(x,\alpha)}{h}\,dx = \int_a^b \frac{\partial f(x,\alpha)}{\partial \alpha}\,dx. \tag{91}$$

**Step 2: Applying the Mean Value Theorem.**

For the moment, hold $x$ fixed and consider $f(x,\alpha)$ as a function of $\alpha$ alone.

Since the partial derivative with respect to $\alpha$ is continuous, we can apply the ordinary Mean Value Theorem.

For sufficiently small $h \neq 0$, there exists a number $\xi_{x,h}$ between $\alpha$ and $\alpha + h$ such that

$$\frac{f(x,\alpha+h)-f(x,\alpha)}{h} = \frac{\partial f(x,\xi_{x,h})}{\partial \alpha}. \tag{92}$$

The subscript $x, h$ reminds us that this intermediate point may depend on both $x$ and $h$.

Since $\xi_{x,h}$ lies between $\alpha$ and $\alpha + h$, we have

$$|\xi_{x,h} - \alpha| \leq |h|. \tag{93}$$

Thus, as $h$ approaches zero, the intermediate point $\xi_{x,h}$ approaches $\alpha$.

By continuity of the partial derivative, we expect

$$\frac{\partial f(x,\xi_{x,h})}{\partial \alpha} \longrightarrow \frac{\partial f(x,\alpha)}{\partial \alpha}.$$

We must now show that this convergence is sufficiently well behaved to pass the limit through the integral.

**Step 3: Estimating the error.**

Let us compare the difference quotient of $F$ with the expression that we want to obtain.

Define the error

$$E(h) = \left| \frac{F(\alpha+h)-F(\alpha)}{h} - \int_a^b \frac{\partial f(x,\alpha)}{\partial \alpha}\,dx \right|. \tag{94}$$

Using equation (90) and the linearity of integration, we obtain

$$E(h) = \left| \int_a^b \left[ \frac{f(x,\alpha+h)-f(x,\alpha)}{h} - \frac{\partial f(x,\alpha)}{\partial \alpha} \right] dx \right|. \tag{95}$$

Now apply the Mean Value Theorem from Step 2:

$$E(h) = \left| \int_a^b \left[ \frac{\partial f(x,\xi_{x,h})}{\partial \alpha} - \frac{\partial f(x,\alpha)}{\partial \alpha} \right] dx \right|. \tag{96}$$

Recall the elementary inequality

$$\left| \int_a^b g(x)\,dx \right| \leq \int_a^b |g(x)|\,dx. \tag{97}$$

Applying this inequality gives

$$E(h) \leq \int_a^b \left| \frac{\partial f(x, \xi_{x,h})}{\partial \alpha} - \frac{\partial f(x, \alpha)}{\partial \alpha} \right| dx. \tag{98}$$

We now want to show that the quantity inside the absolute value becomes uniformly small as $h \to 0$.

**Step 4: Why the error becomes small.**

Since $J$ is open and $\alpha \in J$, we can choose a small number $\delta > 0$ such that

$$[\alpha - \delta, \alpha + \delta] \subset J.$$

Consider the rectangle

$$K = [a, b] \times [\alpha - \delta, \alpha + \delta].$$

This rectangle is closed and bounded, hence compact.

By assumption, the partial derivative

$$\frac{\partial f(x, t)}{\partial \alpha}$$

is continuous on $K$.

A standard result from elementary analysis states that a continuous function on a compact set is uniformly continuous.

In our setting, this means that for every $\varepsilon > 0$, we can choose a sufficiently small $\eta > 0$ such that

$$|t - \alpha| < \eta \quad \Longrightarrow \quad \left| \frac{\partial f(x, t)}{\partial \alpha} - \frac{\partial f(x, \alpha)}{\partial \alpha} \right| < \varepsilon \tag{99}$$

for every $x \in [a, b]$, provided both parameter values lie in the rectangle.

The important point is that we can use the same $\eta$ for every $x \in [a, b]$.

This is precisely what uniform continuity guarantees.

Now recall from (93) that

$$|\xi_{x,h} - \alpha| \leq |h|.$$

Therefore, whenever $h$ is sufficiently small, equation (99) implies

$$\left| \frac{\partial f(x, \xi_{x,h})}{\partial \alpha} - \frac{\partial f(x, \alpha)}{\partial \alpha} \right| < \varepsilon \tag{100}$$

for every $x \in [a, b]$.

Substituting this bound into equation (98), we obtain

$$\begin{aligned} E(h) &\leq \int_a^b \left| \frac{\partial f(x, \xi_{x,h})}{\partial \alpha} - \frac{\partial f(x, \alpha)}{\partial \alpha} \right| dx \\ &\leq \int_a^b \varepsilon \, dx \\ &= \varepsilon(b - a). \end{aligned} \tag{101}$$

Geometrically, this is the area of a rectangle with height $\varepsilon$ and width $(b - a)$.

Thus, we have established

$$\boxed{0 \leq E(h) \leq \varepsilon(b - a)} \tag{102}$$

for all sufficiently small nonzero $h$.

Since $\varepsilon > 0$ can be chosen arbitrarily small and $(b - a)$ is a fixed finite number, the error must approach zero.

More precisely, given any desired error tolerance $\rho > 0$, we may choose

$$\varepsilon = \frac{\rho}{b - a}.$$

Then the preceding inequality gives

$$0 \leq E(h) \leq \rho$$

whenever $h$ is sufficiently small.

Consequently,

$$\lim_{h \to 0} E(h) = 0. \tag{103}$$

**Step 5: Completing the proof.**

By the definition of $E(h)$, the preceding limit means that

$$\lim_{h \to 0} \frac{F(\alpha + h) - F(\alpha)}{h} = \int_a^b \frac{\partial f(x, \alpha)}{\partial \alpha} \, dx.$$

But the expression on the left-hand side is exactly $F'(\alpha)$.

Therefore,

$$\boxed{\frac{d}{d\alpha} \int_a^b f(x, \alpha) \, dx = \int_a^b \frac{\partial f(x, \alpha)}{\partial \alpha} \, dx.} \tag{104}$$

Since $\alpha \in J$ was arbitrary, the result holds throughout $J$.

This completes the proof.

□

The proof reveals the mathematical principle behind Feynman's technique.

### 3.3 When the Limits of Integration Depend on the Parameter

The preceding theorem assumes that the integration limits $a$ and $b$ are fixed.

Suppose instead that the limits are differentiable functions of $\alpha$:

$$F(\alpha) = \int_{a(\alpha)}^{b(\alpha)} f(x, \alpha)\, dx.$$

Under suitable continuity and differentiability assumptions, the general form of Leibniz's rule is

$$\boxed{\begin{aligned} F'(\alpha) = {} & f(b(\alpha), \alpha) b'(\alpha) \\ & - f(a(\alpha), \alpha) a'(\alpha) \\ & + \int_{a(\alpha)}^{b(\alpha)} \frac{\partial f(x, \alpha)}{\partial \alpha}\, dx. \end{aligned}} \tag{105}$$

The first two terms account for the movement of the integration boundaries, while the third term accounts for the change in the integrand itself.

When $a$ and $b$ are constants,

$$a'(\alpha) = b'(\alpha) = 0,$$

and the formula reduces to the familiar identity

$$\frac{d}{d\alpha} \int_a^b f(x, \alpha)\, dx = \int_a^b \frac{\partial f(x, \alpha)}{\partial \alpha}\, dx.$$

This is the special case we have used throughout our examples.

### 3.4 A Remark on Improper Integrals

There is one more subtlety worth mentioning.

Several integrals considered in the preceding section were improper integrals, with an infinite integration interval or a singularity at an endpoint.

The theorem proved above applies directly to a finite interval $[a, b]$. For an improper integral such as

$$F(\alpha) = \int_0^\infty f(x, \alpha)\, dx,$$

additional conditions are required to justify differentiating under the integral sign.

One useful sufficient condition is that, for parameters in a neighborhood of a fixed value $\alpha_0$, there exists an integrable function $g(x)$ such that

$$\left| \frac{\partial f(x, \alpha)}{\partial \alpha} \right| \leq g(x), \tag{106}$$

where

$$\int_0^\infty g(x)\,dx < \infty.$$

If the integrand is differentiable with respect to $\alpha$, the integral defining $F(\alpha)$ is well-defined, and the necessary measurability conditions hold, this domination allows us to justify the interchange using the dominated convergence theorem.

For example, the Gaussian integrands we studied are smooth and decay rapidly at infinity whenever $\alpha > 0$.

Their parameter derivatives can be controlled by integrable functions on suitable neighborhoods of any fixed positive parameter.

The exponentially damped Dirichlet integral admits a similar justification for $\alpha > 0$.

However, recovering the original Dirichlet integral by letting $\alpha \to 0^+$ is a separate limiting question. Its justification cannot be obtained merely by substituting $\alpha = 0$ into the differentiation formula.

These distinctions are important because they show that Feynman's technique is not an unrestricted algebraic manipulation. Its validity depends on the properties of the functions and the domains under consideration.

We have now obtained the mathematical foundation for differentiating under the integral sign.

Nevertheless, another interesting interpretation remains.

Since both integration and differentiation are linear operations, we may ask whether Feynman's technique can be understood in the language of linear operators.

This perspective will be developed in the next section.

# 4 An Operator-Theoretic Interpretation of Feynman's Technique

We have now established the mathematical conditions under which differentiation and integration can be interchanged. However, there is another interesting way to understand Feynman's technique, particularly for readers familiar with linear algebra and matrix mechanics.

Instead of treating differentiation and integration merely as operations from calculus, we can regard them as linear operators acting on spaces of functions.

From this perspective, the central question becomes:

*Under what conditions do the differentiation and integration operators commute?*

We will develop this interpretation step by step.

## 4.1 Functions as Vectors

Recall that a vector space is a collection of objects on which vector addition and scalar multiplication are defined, satisfying the usual vector space axioms.

In elementary linear algebra, we generally work with vectors in $\mathbb{R}^n$.

For example,

$$v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{R}^n.$$

However, vectors need not be finite lists of real numbers.

Functions can also be regarded as vectors.

For example, the collection $C([a,b])$ of continuous real-valued functions on $[a,b]$ forms a vector space under the operations

$$(f+g)(x) = f(x) + g(x)$$

and

$$(cf)(x) = cf(x), \qquad c \in \mathbb{R}.$$

Unlike $\mathbb{R}^n$, this function space is infinite-dimensional.

The important observation is that familiar operations from calculus can be studied as transformations between such vector spaces.

In particular, differentiation and integration are linear transformations.

## 4.2 Differentiation and Integration as Linear Operators

Let us first consider the differentiation operator

$$Df = \frac{df}{dx}.$$

For differentiable functions $f$ and $g$, and constants $c_1, c_2$, we have

$$\begin{aligned} D(c_1 f + c_2 g) &= \frac{d}{dx}(c_1 f + c_2 g) \\ &= c_1 \frac{df}{dx} + c_2 \frac{dg}{dx} \\ &= c_1 Df + c_2 Dg. \end{aligned} \tag{107}$$

Thus, differentiation is a linear operator.

Similarly, define the integration operator

$$Tf = \int_a^b f(x)\,dx. \tag{108}$$

By linearity of integration,

$$
\begin{aligned}
T(c_1 f + c_2 g) &= \int_a^b (c_1 f(x) + c_2 g(x))\, dx \\
&= c_1 \int_a^b f(x)\, dx + c_2 \int_a^b g(x)\, dx \\
&= c_1 T f + c_2 T g.
\end{aligned} \tag{109}
$$

Therefore, integration is also a linear operator.

Notice that differentiation and definite integration need not have the same domain and codomain.

For example, differentiation maps differentiable functions to functions, whereas definite integration over a fixed interval maps functions to real numbers.

When a parameter is present, the situation becomes particularly interesting.

## 4.3 Introducing the Parameter

Suppose that we have a function $f(x, \alpha)$, where $x$ is the variable of integration and $\alpha$ is a parameter.

Let $J$ be an open interval containing the allowed values of $\alpha$.

We define the integration operator with respect to $x$ by

$$
(T_x f)(\alpha) = \int_a^b f(x, \alpha)\, dx. \tag{110}
$$

Notice that $T_x f$ is now a function of $\alpha$.

Integration has eliminated the dependence on $x$, but the parameter remains.

Next, define differentiation with respect to $\alpha$:

$$
(D_\alpha f)(x, \alpha) = \frac{\partial f(x, \alpha)}{\partial \alpha}. \tag{111}
$$

We also use $D_\alpha$ to denote ordinary differentiation when acting on a function of $\alpha$ alone.

Both operations are linear, but they act in different ways.

The integration operator $T_x$ integrates over the variable $x$, whereas $D_\alpha$ measures variation with respect to the parameter $\alpha$.

Our objective is to investigate whether the order of these operations matters.

## 4.4 Commutativity of the Operators

Consider first applying the integration operator and then differentiating the result.

We obtain

$$
D_\alpha (T_x f) = \frac{d}{d\alpha} \left( \int_a^b f(x, \alpha)\, dx \right). \tag{112}
$$

Now reverse the order.

First differentiate the integrand with respect to $\alpha$, and then integrate over $x$:

$$T_x(D_\alpha f) = \int_a^b \frac{\partial f(x,\alpha)}{\partial \alpha}\, dx. \tag{113}$$

Feynman's technique asks whether these two expressions are equal.

In other words,

$$\boxed{D_\alpha(T_x f) = T_x(D_\alpha f).} \tag{114}$$

But this is exactly the statement of Leibniz's integral rule for fixed integration limits.

Consequently, under the conditions established in the preceding section, the two operations can be performed in either order without changing the result.

This provides an operator-theoretic interpretation of differentiation under the integral sign.

**The commutator**

In linear algebra and operator theory, the *commutator* of two operators $A$ and $B$ acting on a common vector space is defined by

$$[A, B] = AB - BA. \tag{115}$$

The operators commute precisely when

$$[A, B] = 0.$$

We would therefore like to express Feynman's technique in the form

$$[D_\alpha, T_x] = 0.$$

There is a small technical point here.

The integration operator defined in (110) maps a function of $(x, \alpha)$ to a function of $\alpha$ alone.

Strictly speaking, $D_\alpha$ and $T_x$ are therefore not both endomorphisms of the same function space.

The precise mathematical statement is the identity (114), where differentiation acts on the appropriate space before or after integration.

If we wish to use the ordinary commutator notation, we can regard the output of $T_x$ as a function that is constant in $x$.

More explicitly, define

$$(\widetilde{T}_x f)(x,\alpha) = \int_a^b f(t,\alpha)\, dt. \tag{116}$$

Here $t$ is merely a dummy integration variable.

Unlike the original integration operator, $\widetilde{T}_x$ maps functions of $(x, \alpha)$ back into functions of $(x, \alpha)$.

We can now form the commutator in the usual sense:

$$[D_\alpha, \widetilde{T}_x] = D_\alpha \widetilde{T}_x - \widetilde{T}_x D_\alpha. \tag{117}$$

Under the hypotheses of Leibniz's rule,

$$\boxed{[D_\alpha, \widetilde{T}_x] f = 0.} \tag{118}$$

Thus, on a suitable space of functions, the differentiation and fixed-interval integration operators commute.

This is the operator identity underlying Feynman's technique.

### 4.5 A Finite-Dimensional Analogy

We can make the preceding discussion more intuitive by considering an analogous problem in ordinary linear algebra.

Suppose that

$$v(\alpha) = \begin{pmatrix} v_1(\alpha) \\ v_2(\alpha) \\ \vdots \\ v_n(\alpha) \end{pmatrix}$$

is a differentiable vector-valued function.

Let

$$w = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}$$

be a fixed vector, independent of $\alpha$.

Consider the scalar-valued function

$$F(\alpha) = w^T v(\alpha).$$

By the definition of matrix multiplication,

$$F(\alpha) = \sum_{i=1}^{n} w_i v_i(\alpha).$$

Differentiating with respect to $\alpha$,

$$
\begin{aligned}
F'(\alpha) &= \frac{d}{d\alpha} \sum_{i=1}^{n} w_i v_i(\alpha) \\
&= \sum_{i=1}^{n} w_i v_i'(\alpha) \\
&= w^T v'(\alpha).
\end{aligned} \tag{119}
$$

Therefore,

$$
\boxed{\frac{d}{d\alpha}(w^T v(\alpha)) = w^T v'(\alpha).} \tag{120}
$$

Notice that differentiation can be performed before or after taking the weighted sum.

Now recall that a definite integral can be approximated by a Riemann sum:

$$
\int_a^b f(x, \alpha)\, dx \approx \sum_{i=1}^{n} f(\xi_i, \alpha) \Delta x_i.
$$

This is precisely a weighted sum of the form

$$
w^T v(\alpha),
$$

where

$$
w_i = \Delta x_i, \qquad v_i(\alpha) = f(\xi_i, \alpha).
$$

Since the weights $\Delta x_i$ are independent of $\alpha$, we can differentiate the finite sum componentwise.

The passage from finite sums to integrals is what makes Feynman's technique a natural infinite-dimensional analogue of (120).

However, unlike a finite sum, an integral involves a limiting process.

Consequently, additional regularity conditions are needed to justify the interchange in the infinite-dimensional case.

### 4.6 Why Linearity Alone Is Not Sufficient

It is important to emphasize that the vanishing commutator does not follow merely from the linearity of the two operators.

Indeed, even in ordinary linear algebra, two matrices $A$ and $B$ may both represent linear transformations without commuting:

$$
AB \neq BA.
$$

Similarly, differentiation and integration are linear operations, but this fact alone does not establish that their order can always be interchanged.

The equality

$$D_\alpha T_x f = T_x D_\alpha f$$

must be justified by the properties of the function $f$ and the domain on which the operators act.

For continuously differentiable integrands over fixed finite intervals, Leibniz's rule provides that justification.

For improper integrals, additional convergence conditions are required.

Furthermore, if the integration limits depend on the parameter, then Leibniz's general rule gives

$$\begin{aligned}
&\frac{d}{d\alpha}\int_{a(\alpha)}^{b(\alpha)} f(x,\alpha)\,dx \\
&\quad - \int_{a(\alpha)}^{b(\alpha)} \frac{\partial f(x,\alpha)}{\partial \alpha}\,dx \\
&= f(b(\alpha),\alpha)b'(\alpha) - f(a(\alpha),\alpha)a'(\alpha).
\end{aligned} \tag{121}$$

Thus, additional boundary terms appear when the limits move with the parameter.

The operator identity therefore has a precise domain of validity, and one should be careful before exploiting it in scientific computing.

### 4.7 Returning to Feynman's Technique

Let us briefly revisit the Gaussian integral from the operator perspective.

Recall that

$$G(\alpha) = \int_0^\infty e^{-\alpha x^2}\,dx = \frac{\sqrt{\pi}}{2\sqrt{\alpha}}, \qquad \alpha > 0. \tag{122}$$

Differentiating the explicit expression,

$$G'(\alpha) = -\frac{\sqrt{\pi}}{4\alpha^{3/2}}. \tag{123}$$

On the other hand, differentiating under the integral sign,

$$\begin{aligned}
G'(\alpha) &= \int_0^\infty \frac{\partial}{\partial \alpha} e^{-\alpha x^2}\,dx \\
&= -\int_0^\infty x^2 e^{-\alpha x^2}\,dx.
\end{aligned} \tag{124}$$

Equating the two expressions gives

$$\int_0^\infty x^2 e^{-\alpha x^2}\,dx = \frac{\sqrt{\pi}}{4\alpha^{3/2}}.$$

At $\alpha = 1$, we obtain

$$\boxed{\int_0^\infty x^2 e^{-x^2}\,dx = \frac{\sqrt{\pi}}{4}.} \tag{125}$$

In operator notation, the essential step was

$$D_\alpha(T_x f) = T_x(D_\alpha f),$$

where

$$f(x,\alpha) = e^{-\alpha x^2}.$$

The two operators produced the same result, but one order of calculation was much more useful than the other.

This is precisely the computational advantage exploited by Feynman's technique.

# 5 Conclusion

In this note, we have explored Feynman's technique of parametric integration, beginning with its computational applications and concluding with its mathematical justification.

We first considered several examples involving Gaussian, trigonometric, and logarithmic integrals. In each case, the essential idea was to introduce or exploit a parameter whose differentiation simplifies the original integrand or leads to a differential equation satisfied by the parametric integral itself.

The Gaussian integrals illustrated how repeated differentiation generates powers of the integration variable and naturally leads to ordinary and double factorials. The Dirichlet integral demonstrated how exponential damping can transform a difficult oscillatory integral into an elementary one. The Gaussian–cosine integral showed how differentiation under the integral sign, combined with integration by parts, can transform an integration problem into a Cauchy problem, which can then be solved using a known initial condition. Finally, the logarithmic integral showed how differentiation can eliminate an otherwise troublesome logarithmic denominator.

Having developed the computational method, we investigated why differentiation and integration can sometimes be interchanged. Starting from the Riemann-sum interpretation of integration, we were led to Leibniz's integral rule, which provides a rigorous mathematical foundation for the technique.

We then examined the method from the perspective of linear algebra and operator theory. By regarding differentiation and integration as linear operators acting on suitable spaces of functions, we interpreted Leibniz's rule as a commutation relation.

In particular, under appropriate regularity and convergence conditions, we have

$$D_\alpha T_x f = T_x D_\alpha f. \tag{126}$$

Equivalently, using the extended integration operator introduced earlier, the corresponding commutator vanishes:

$$\boxed{[D_\alpha, \widetilde{T}_x]f = 0.} \tag{127}$$

It is important to remember, however, that this relation does not hold for arbitrary functions. Its validity depends on the mathematical conditions governing the interchange of differentiation and integration.

The broader lesson is that many difficult mathematical problems become simpler when we recognize the underlying structure of the operations involved. Rather than evaluating an integral directly, Feynman's technique allows us to embed it in a family of integrals whose dependence on a parameter is easier to investigate.

Thus, what initially appears to be a clever computational trick is, in fact, an application of a deeper principle from mathematical analysis and operator theory.

The technique illustrates an important theme in mathematics: understanding when and why operations can be interchanged is often just as valuable as knowing how to perform them.

## References


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